

### Question 1: Basic TRY...CATCH for Error Logging

```
CREATE PROCEDURE AddEmployee
AS
BEGIN
BEGIN TRY
INSERT INTO Employees VALUES
(@ID,@First,@Last,@Email,@Salary,@Dept);
END TRY
BEGIN CATCH
INSERT INTO AuditLog(Action,ErrorMessage)
VALUES('Insert Error',ERROR_MESSAGE());
END CATCH
END;
```

|                             |
|-----------------------------|
| Result                      |
| Employee Added Successfully |
| Error Logged                |

### Question 2: Using THROW

```
BEGIN CATCH
INSERT INTO AuditLog(Action,ErrorMessage)
VALUES('Error',ERROR_MESSAGE());
THROW;
END CATCH;
```

|                            |
|----------------------------|
| Message                    |
| Error Logged and Re-thrown |

### Question 3: Custom Error with RAISERROR

```
IF @Salary <= 0
BEGIN
RAISERROR
('Salary must be greater than zero.',16,1);
RETURN;
END
```

|                                   |
|-----------------------------------|
| Message                           |
| Salary must be greater than zero. |

#### ***Question 4: Nested TRY...CATCH***

```
BEGIN TRY
UPDATE Employees
SET DepartmentID=@Dept;
END TRY
BEGIN CATCH
INSERT INTO AuditLog(Action,ErrorMessage)
VALUES('Transfer Error',ERROR_MESSAGE());
THROW;
END CATCH
```

|                         |
|-------------------------|
| Message                 |
| Transaction Rolled Back |

#### ***Question 5: Transaction Error Logging***

```
BEGIN TRANSACTION
BEGIN TRY
INSERT INTO Employees VALUES(...);
COMMIT;
END TRY
BEGIN CATCH
ROLLBACK;
INSERT INTO AuditLog(Action,ErrorMessage)
VALUES('Batch Error',ERROR_MESSAGE());
END CATCH
```

|                     |
|---------------------|
| Message             |
| Batch Insert Failed |

### Question 6: Dynamic RAISERROR

```
IF @Salary < 1000  
RAISERROR('Salary Warning',10,1);
```

```
IF @Salary < 0  
RAISERROR('Invalid Salary Error',16,1);
```

| Severity | Message              |
|----------|----------------------|
| 10       | Salary Warning       |
| 16       | Invalid Salary Error |